

# Savior Glass / RoboEye

## Progress and publication-status report

Date: 21 September 2026

Project: Wearable assistive smart glass (OCR, objects, Bangladeshi Taka, emotion-adaptive speech)

Audience: supervisor / collaborator

Rule: every accuracy figure below was measured in this workspace.

**VERDICT: Not eligible for an A\* conference (CVPR, ICCV, NeurIPS, ICLR, CHI) as the work stands.**

**Eligible: thesis, live demo, IEEE application / regional conference.**

**Emotion (official RAF-DB test): 86.5% | requested 93-96%: not reached.**

### 1. Executive summary

This is a working senior-project prototype that fine-tunes existing models. It is not a new computer-vision method and it has no controlled user study with visually impaired participants.

- Emotion on official RAF-DB test (3,068 images): 86.5% accuracy, macro-F1 0.80. Was 81.0% before this week's GPU training.
- Published 7-class RAF-DB SOTA is about 92.2% (POSTER V2). 93-96% sits above typical published numbers on this split.
- Taka on 450 wild notes: 98.4% detection, 96.9% top-1 denomination (YOLOv8s fine-tune).
- Do not put 93-96% emotion or 96.7% authenticity into any slide, email, or draft.

### 2. What was implemented

| Area          | What landed                                      | Status                             |
|---------------|--------------------------------------------------|------------------------------------|
| Emotion       | EfficientNet-V2-S + FER+ fallback + adaptive TTS | Working, 86.5% RAF-DB              |
| Currency      | YOLOv8s 9-class Taka; ONNX/INT8                  | Working                            |
| Authenticity  | CNN+ViT, dual-head, FedAvg, EWC                  | Code yes; acc 1.00 not publishable |
| OCR / objects | EasyOCR BN+EN; YOLOv8 COCO                       | Working                            |
| Assistive I/O | TTS, ASR, haptics, pose, Grad-CAM                | PC working; not bone-conduction/VR |

### 3. Emotion experiments this week

GPU: RTX 3050 Laptop after installing CUDA PyTorch (it had been CPU-only). Dataset already on disk: RAF-DB train 12,271 / official test 3,068. Live weights: savior\_glass/models/emotion\_faces.pt

| Run                        | Test            | Acc        | Kept?      |
|----------------------------|-----------------|------------|------------|
| FER+ ONNX (older)          | FER-2013 n=5741 | 57.9%      | Fallback   |
| HGB + FER+ blend           | FER-2013 n=5741 | 67.6%      | No         |
| MobileNetV3-Large (before) | RAF-DB official | 81.0%      | Replaced   |
| EfficientNet-V2-S + TTA    | RAF-DB official | 86.5%      | YES - live |
| ViT fine-tune              | RAF-DB official | 84.3% peak | Stopped    |

|                      |                 |                    |         |
|----------------------|-----------------|--------------------|---------|
| AffectNet ENET-B2 FT | RAF-DB official | 85.2% then overfit | Stopped |
|----------------------|-----------------|--------------------|---------|

Best model per class (RAF-DB official test)

| Class    | n    | Prec          | Rec           | F1               |
|----------|------|---------------|---------------|------------------|
| Happy    | 1185 | 0.946         | 0.925         | 0.936            |
| Sad      | 478  | 0.839         | 0.839         | 0.839            |
| Angry    | 162  | 0.789         | 0.809         | 0.799            |
| Surprise | 329  | 0.867         | 0.872         | 0.870            |
| Fear     | 74   | 0.774         | 0.554         | 0.646            |
| Disgust  | 160  | 0.671         | 0.588         | 0.627            |
| Neutral  | 680  | 0.815         | 0.890         | 0.851            |
| OVERALL  | 3068 | macro-P 0.815 | macro-R 0.782 | 86.5% / F1 0.795 |

Bottleneck: Fear (74 images) and Disgust (160). Happy is already strong (F1 0.936).

4. Other measured results

| Task          | Protocol                         | Result                           |
|---------------|----------------------------------|----------------------------------|
| Taka wild     | 450 images, YOLOv8s best.pt      | Detect 98.4%, top-1 96.9%        |
| Taka YOLO val | Saved val (synthetic composites) | mAP50 0.994, mAP50-95 0.844      |
| OCR           | EasyOCR n=80 (40 BN + 40 EN)     | CER 15.5% (BN 27.8%, EN 3.2%)    |
| Objects       | YOLOv8s, 250 COCO val subset     | mAP50 0.60                       |
| Authenticity  | Local JaalTaka split             | Acc 1.00 - LEAKY, do not publish |

5. Why this is not A\* material

A\* reviewers expect a new method, a SOTA (or large clean) gain on a standard benchmark, or a CHI/ASSETS-grade user study. This repo fine-tunes public models for a wearable pipeline.

- CVPR / ICCV / ECCV: No. Emotion 86.5% is below ~92% SOTA; Taka is a YOLOv8s application.
- NeurIPS / ICML / ICLR: No. There is no new learning algorithm.
- CHI / ASSETS: Not yet. Need a study with blind/low-vision users, not only a PC demo.

Reasonable venues now: thesis examination, system demo, IEEE application or regional conferences (TENCON, ICCIT, ICAEE, similar). A later accessibility paper still needs a user study.

6. Do not claim

- 93-96% emotion (not measured).
- 96.7% / 98.1% authenticity (never measured by this stack).
- Raspberry Pi 15 FPS (not timed on a Pi).
- Qwen-VL / LLaVA / bone-conduction hardware (not what runs today).
- Full 22,315-image composite set on this machine (script exists; set not generated here).

## 7. What to do now

Freeze 86.5% emotion and present a thesis + live demo. Record 1-2 minutes: note detection, speech, emotion-adaptive TTS. Only restart emotion training if the goal is a higher honest RAF-DB number (for example POSTER V2), and report that number even if it is 90-92%.

## 8. Key files

```
savior_glass/models/emotion_faces.pt  
savior_glass/results/emotion_rafdb.json  
savior_glass/scripts/train_emotion_rafdb.py  
realtime_bangla_taka_detection/roboeye/fer_emotion.py  
realtime_bangla_taka_detection/models/best.pt
```

*Prepared from local evaluation JSON in this workspace, 21 September 2026.*